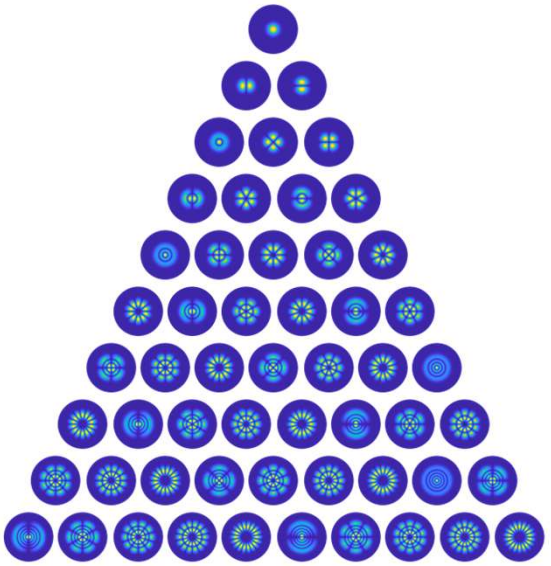
University of Stuttgart
Institute of electrical and optical communications

# Advanced optical fiber communications

## Wintersemester 2025/26

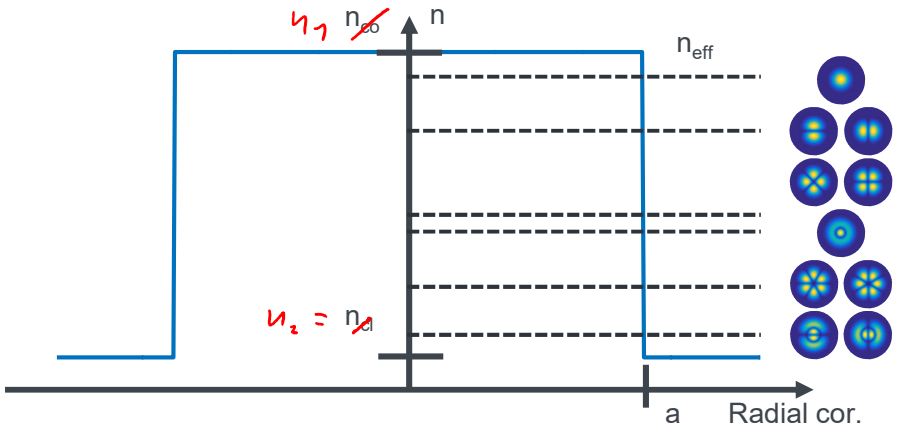
## Lecture 7

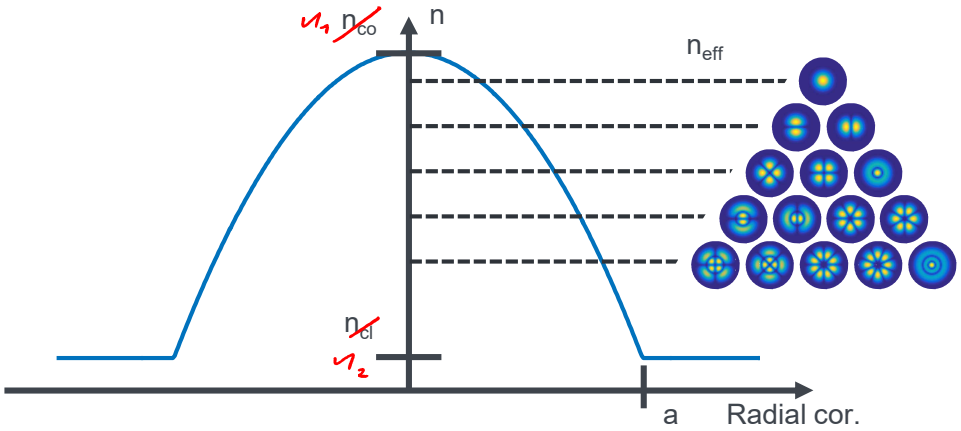


Prof. Dr.-Ing. Georg Rademacher

# Multi-mode fibers (MMFs)

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Germany



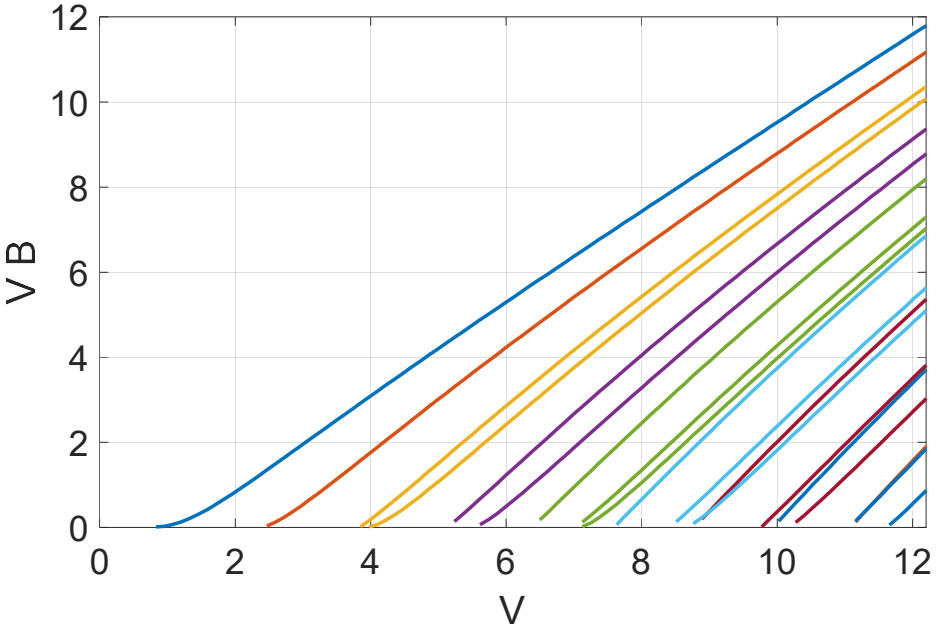



# Graded-index fibers

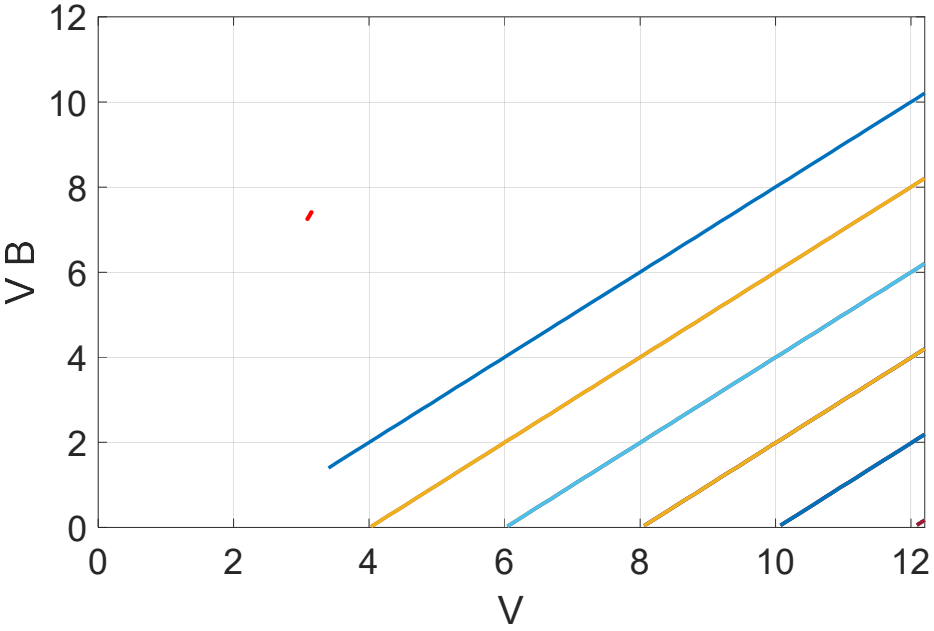
University of Stuttgart
Germany

$$\tau = \frac{n_1 - n_2}{c} \cdot \frac{d(V \cdot B)}{dV} + \frac{1}{c} N_2$$

SI-Fiber



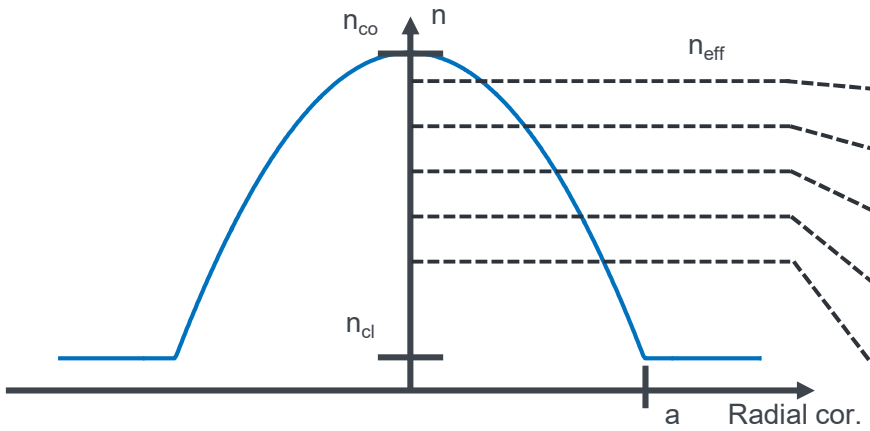

GI-Fiber

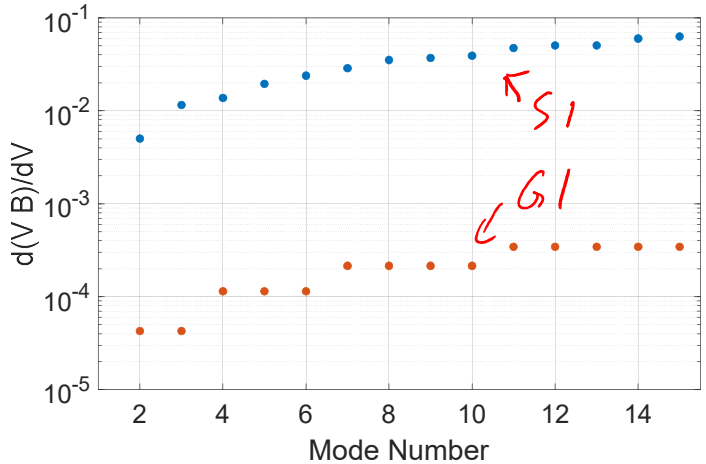


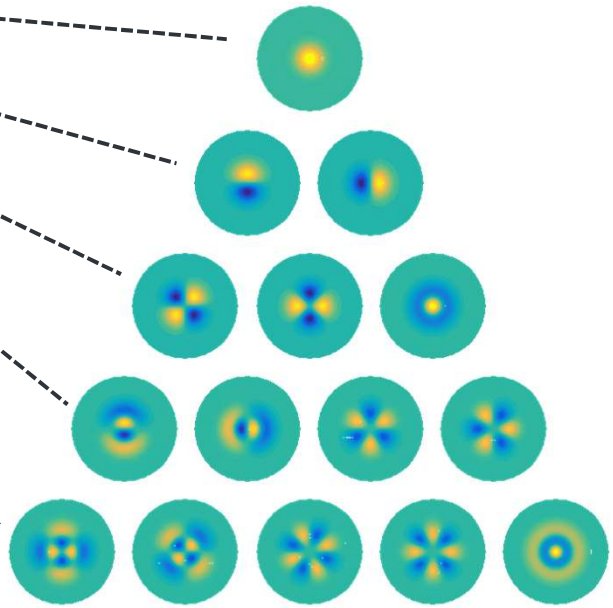
# Multi-mode fibers (MMFs)

- Nomenklatur!

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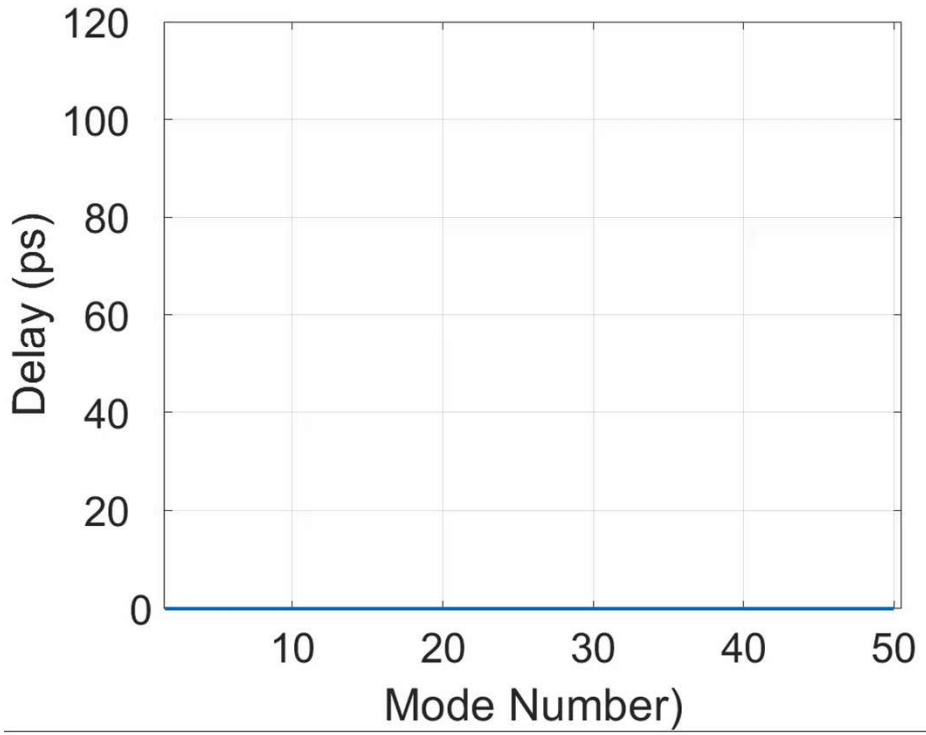



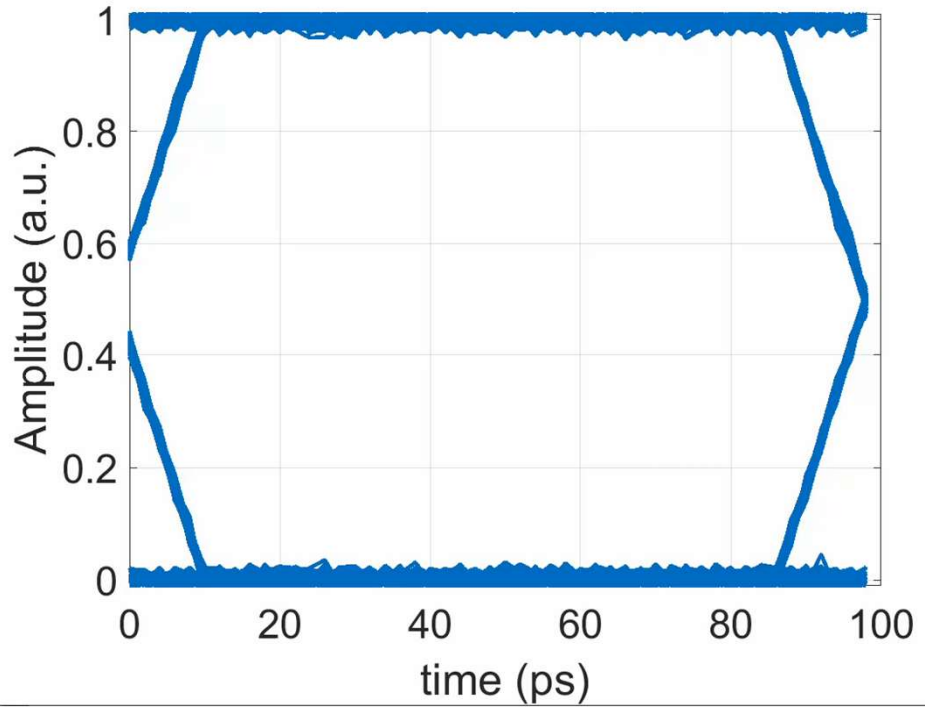



$$\beta_{LP} = n_1 k_0 \left(1 - \frac{\sqrt{2\Delta}}{n_1 k_0 a}(2L+4P-2)\right)^{1/2}$$

# Why is the delay difference so important?

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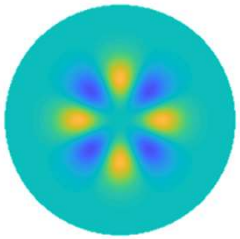
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# Bonus Test

## Bonus test

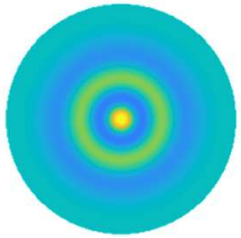
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Germany

Which mode is in the same group as the shown profile of LP41?

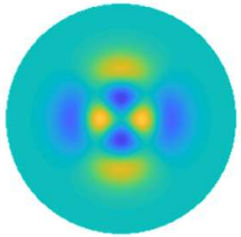


$2 \cdot L + 4 \quad P = 12$

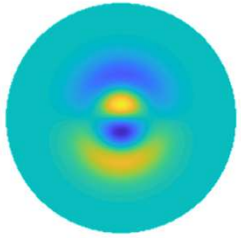
A



B



C



https://ilias3.uni-stuttgart.de/goto.php?target=xlvo_1_pin_OXCT

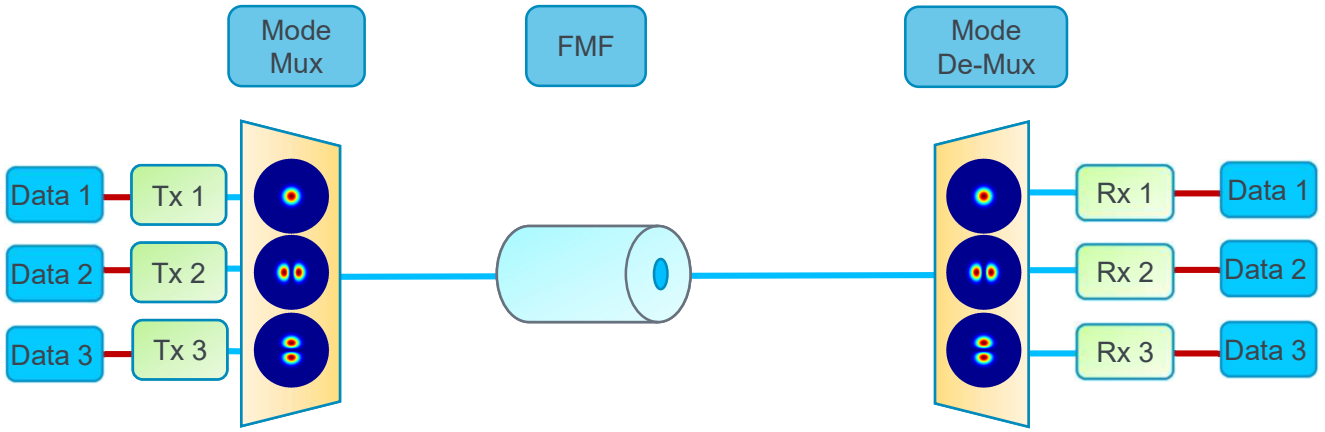
# LV-evaluation

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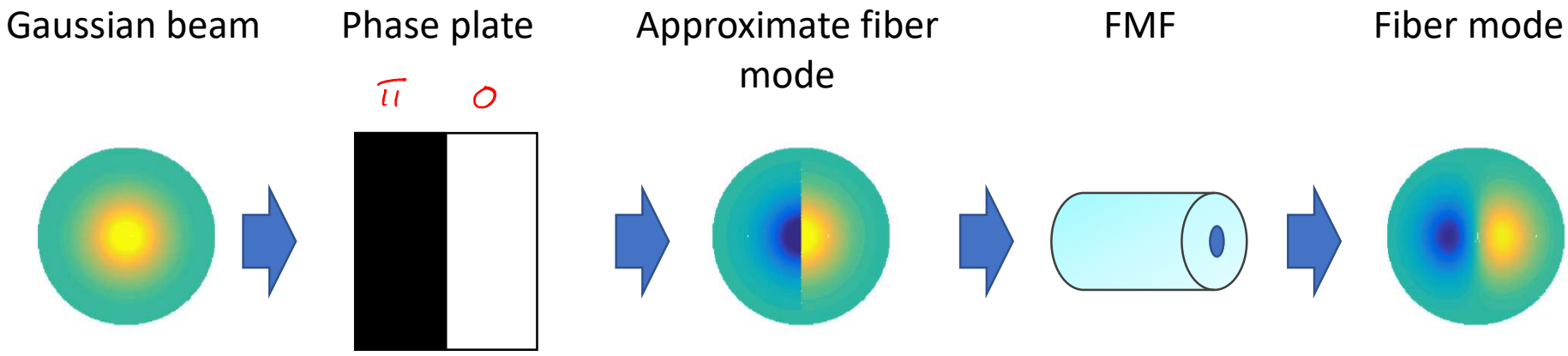
# Space division multiplex with MMF

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# Mode-Multiplexers

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- Draw how to make 3-mode device with phase plates, mirrors and beam splitter

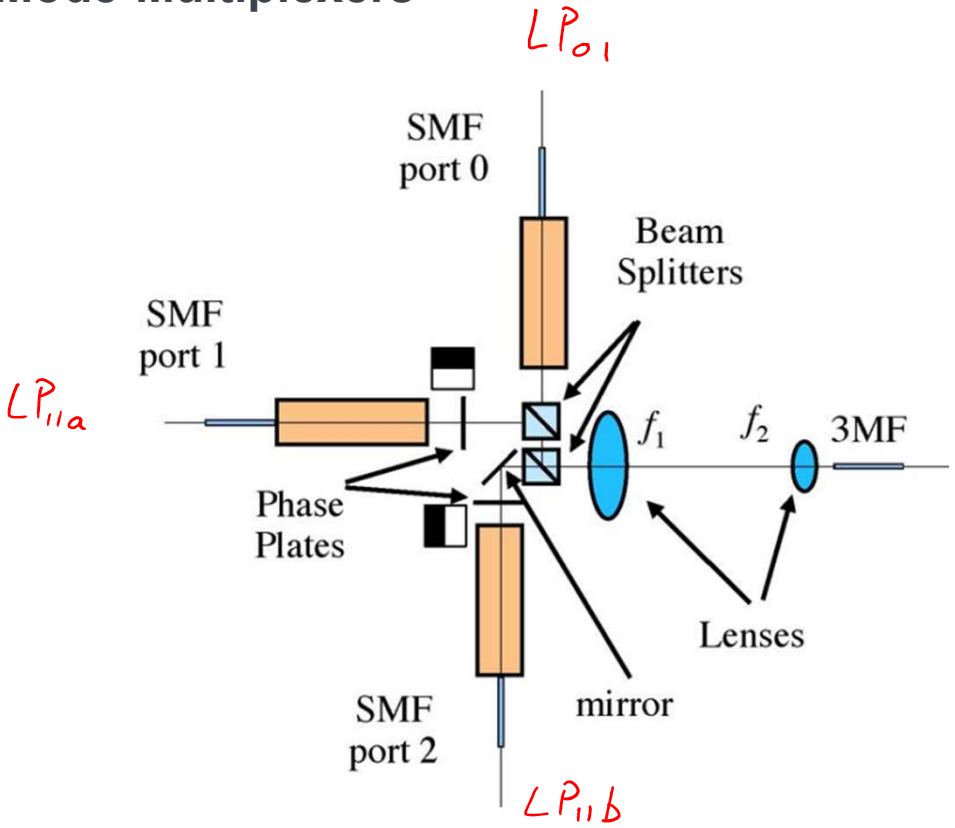
JOURNAL OF LIGHTWAVE TECHNOLOGY, VOL. 30, NO. 4, FEBRUARY 15, 2012 

# Mode-Division Multiplexing Over 96 km of Few-Mode Fiber Using Coherent $6 \times 6$ MIMO Processing

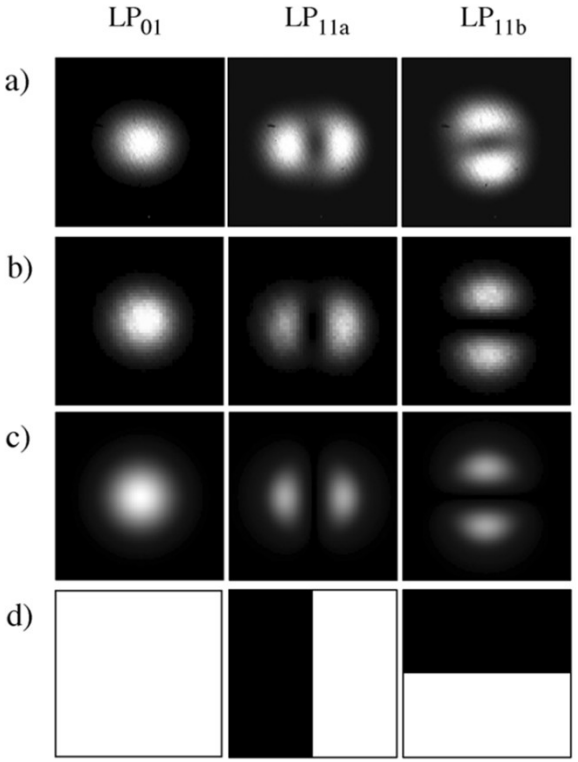
Roland Ryf, *Member, IEEE*, Sebastian Randel, *Senior Member, IEEE*, Alan H. Gnauck, *Fellow, IEEE, Fellow, OSA*, Cristian Bolle, Alberto Sierra, *Student Member, IEEE*, Sami Mumtaz, Mina Esmaeelpour, Ellsworth C. Burrows, René-Jean Essiambre, *Senior Member, IEEE, Fellow, OSA*, Peter J. Winzer, *Fellow, IEEE, Member, OSA*, David W. Peckham, *Senior Member, IEEE*, Alan H. McCurdy, *Member, IEEE*, and Robert Lingle, Jr., *Member, IEEE*

*(Invited Tutorial)*

# Mode-Multiplexers




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ARTICLE

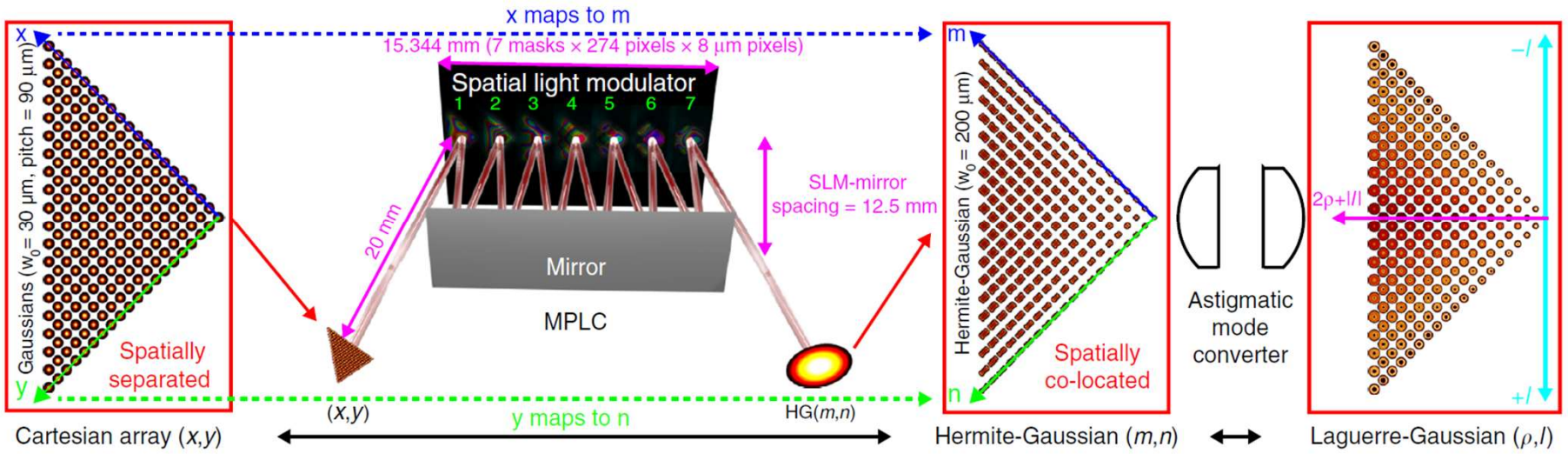
https://doi.org/10.1038/s41467-019-09840-4 OPEN

# Laguerre-Gaussian mode sorter

Nicolas K. Fontaine[1,3], Roland Ryf[1], Haoshuo Chen[1], David T. Neilson[1], Kwangwoong Kim[1] & Joel Carpenter[2,3]

# Multi-plane light conversion

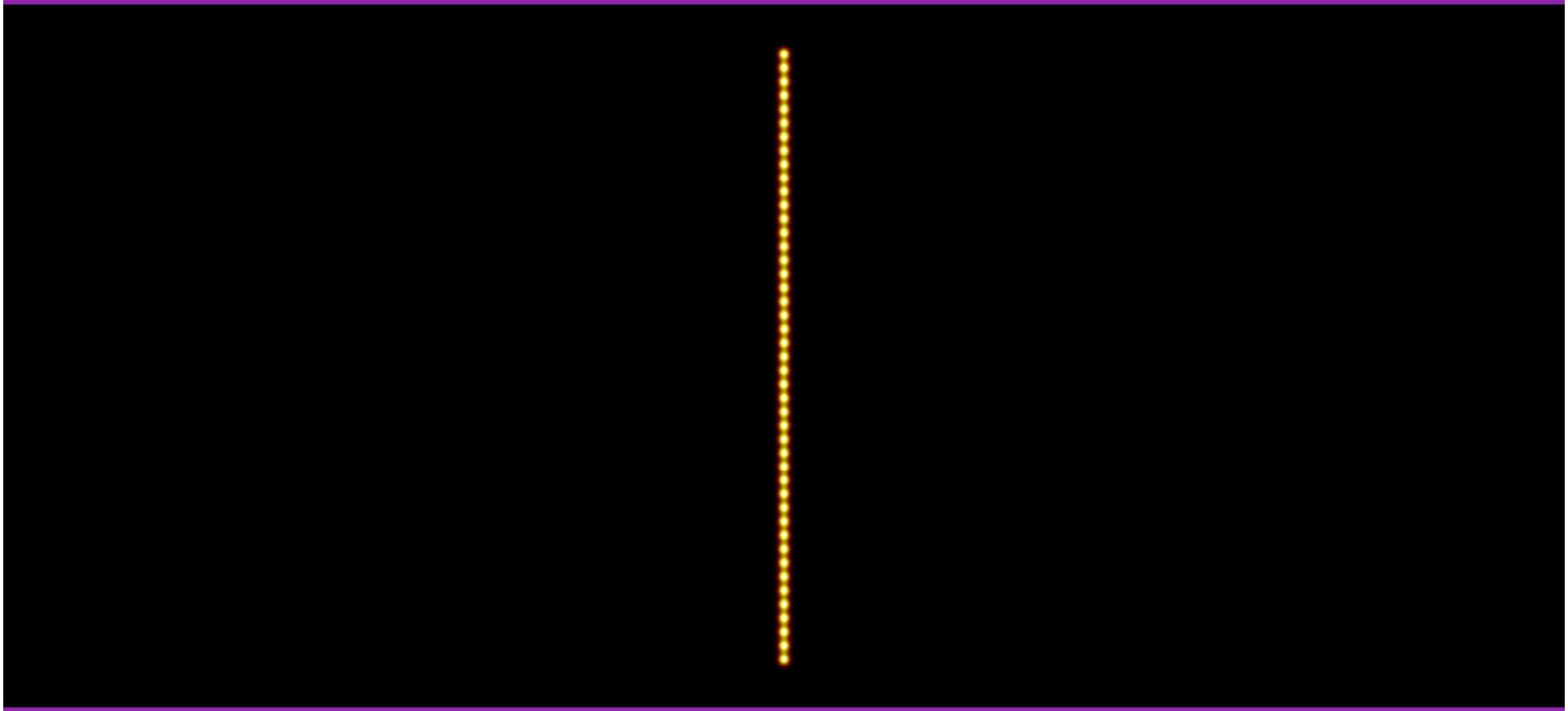
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AUSTRALIA

# Multi-plane light conversion

NOKIA Bell Labs

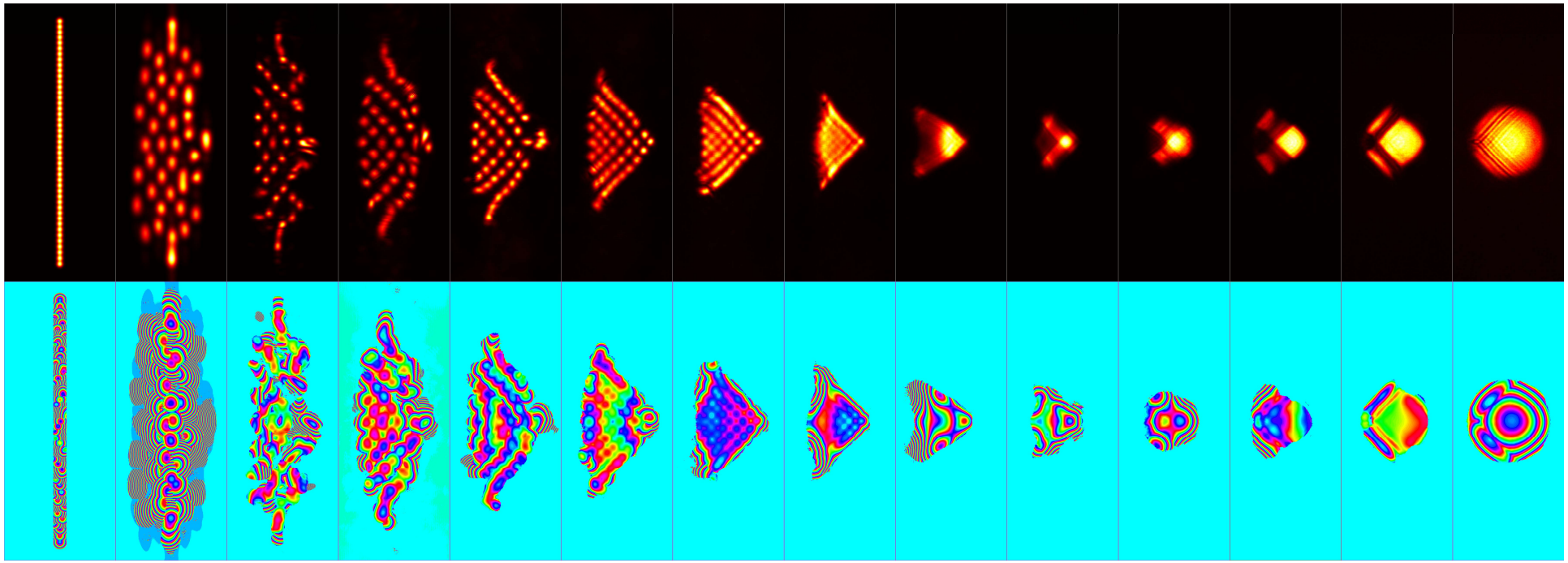


J.-F Morizur et al., J. Opt. Soc. Am. A, 27 (11) 2524-2531 (2010).

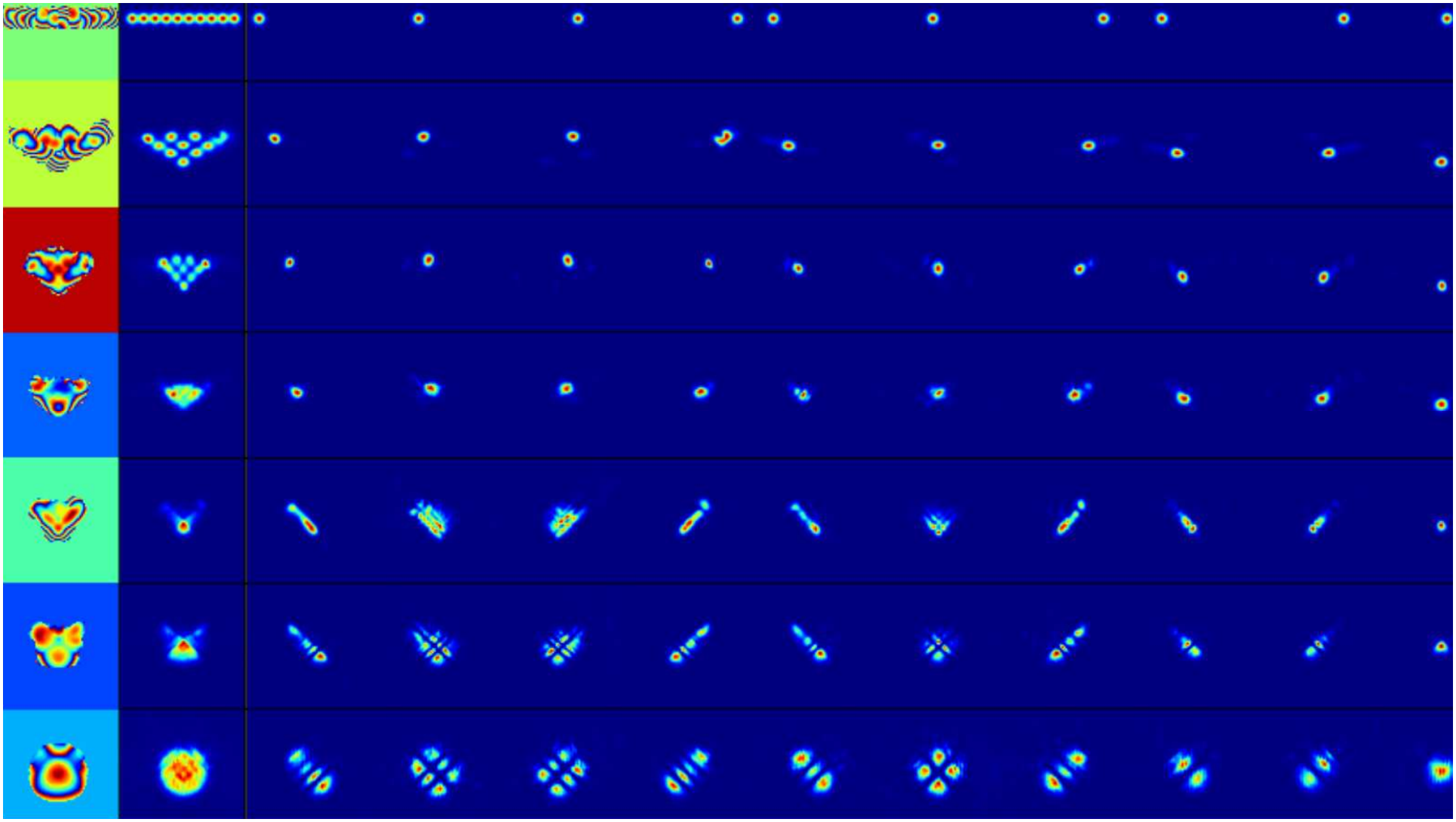
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# Triangular → HG

NOKIA Bell Labs

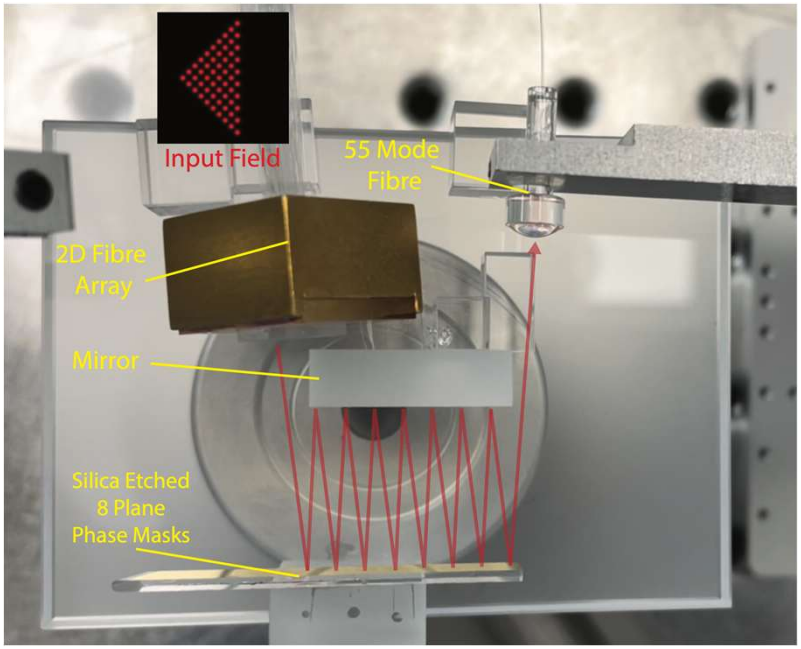


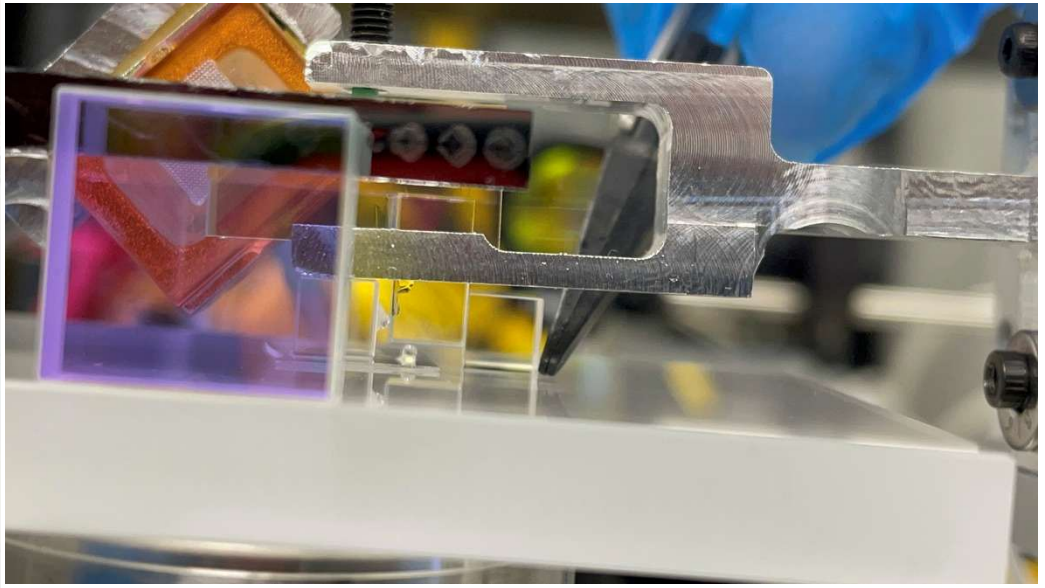
45 modes, 14 planes (512×1280 5μm pixels @ 1550nm, 24mm plane spacing)



# Mode-Multiplexers

University of Stuttgart
Germany

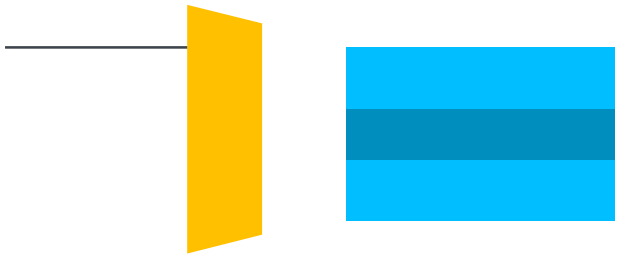


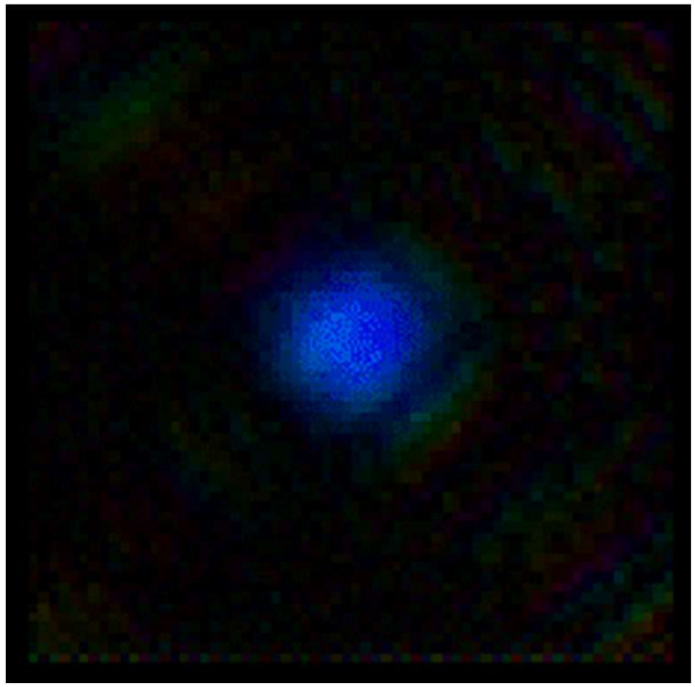



# Mode-Multiplexers

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- SMF
- MUX
- MMF





# Mode-Multiplexers

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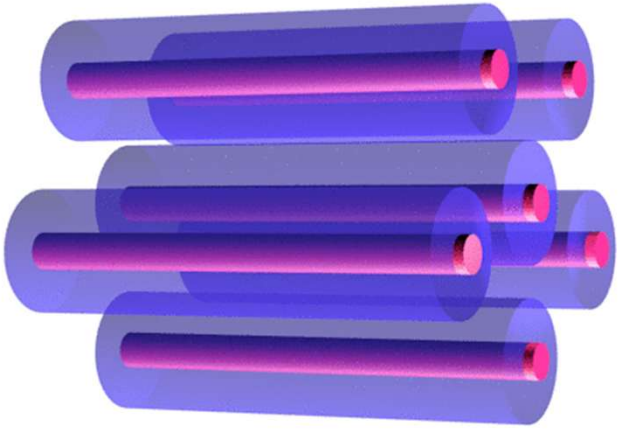

## Photonic Lanterns, 3-D Waveguides, Multiplane Light Conversion, and Other Components That Enable Space-Division Multiplexing

*This article reviews the mode-shaping technologies needed to perform mode-division multiplexing on few-mode fibers and discusses implementations and tradeoffs of four different techniques: multiplane light conversion, fused fiber devices, 3-D waveguides in glass, and free-space imaging systems.*

By Nicolas K. Fontaine, *Senior Member IEEE*, Joel Carpenter, *Member IEEE*, Simon Gross, Sergio Leon-Saval, Yongmin Jung, David J. Richardson, *Fellow IEEE*, and Rodrigo Amezcua-Correa

# Mode-Multiplexers: Photonic Lantern





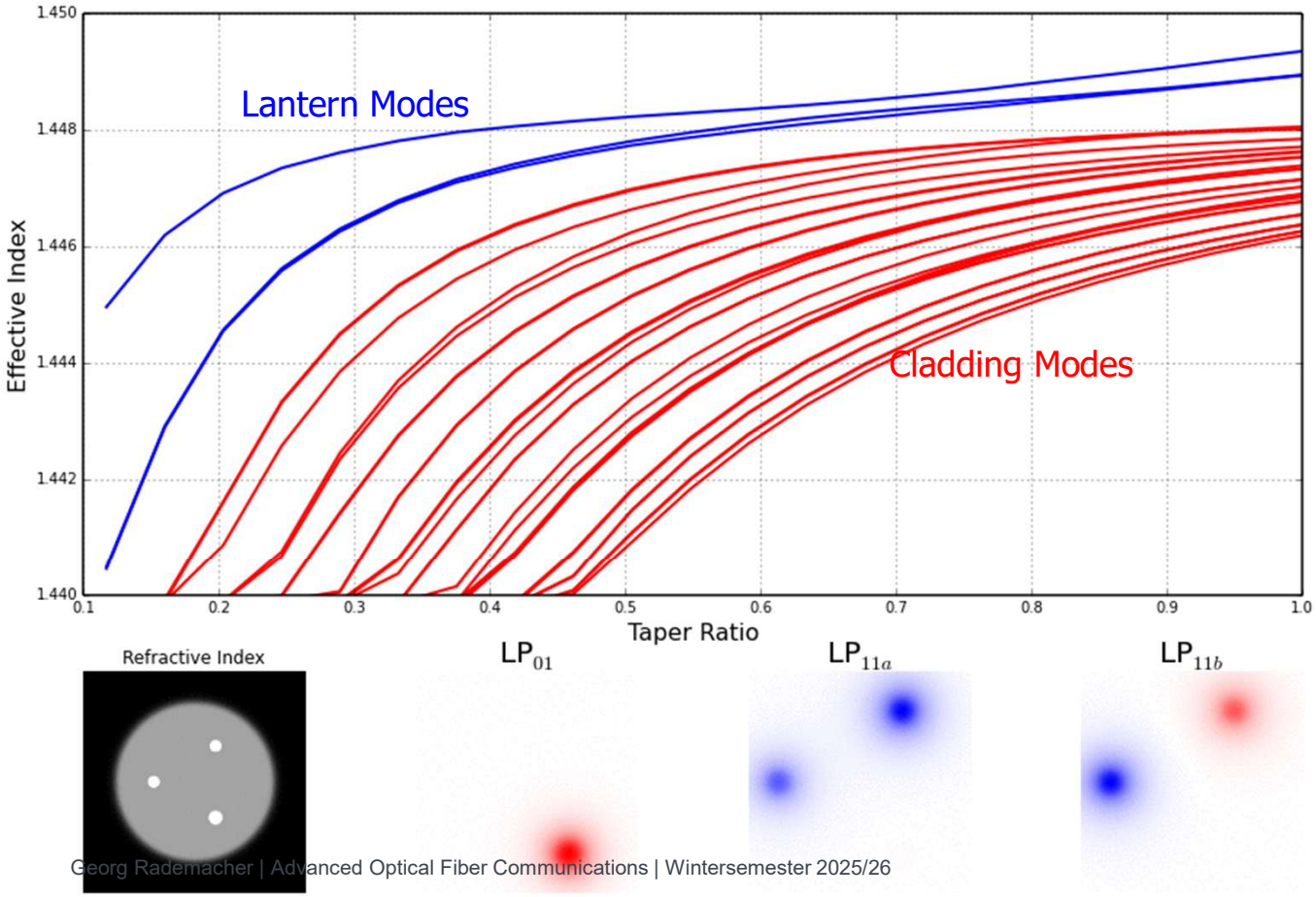


## issimilar Fiber Mode Selective Lantern
### diabatic transition from groups of cores to modes.

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rmany

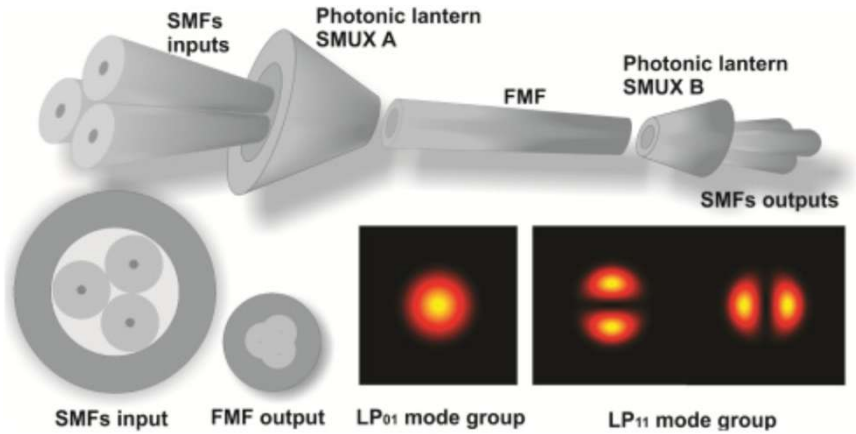
01 Big Core

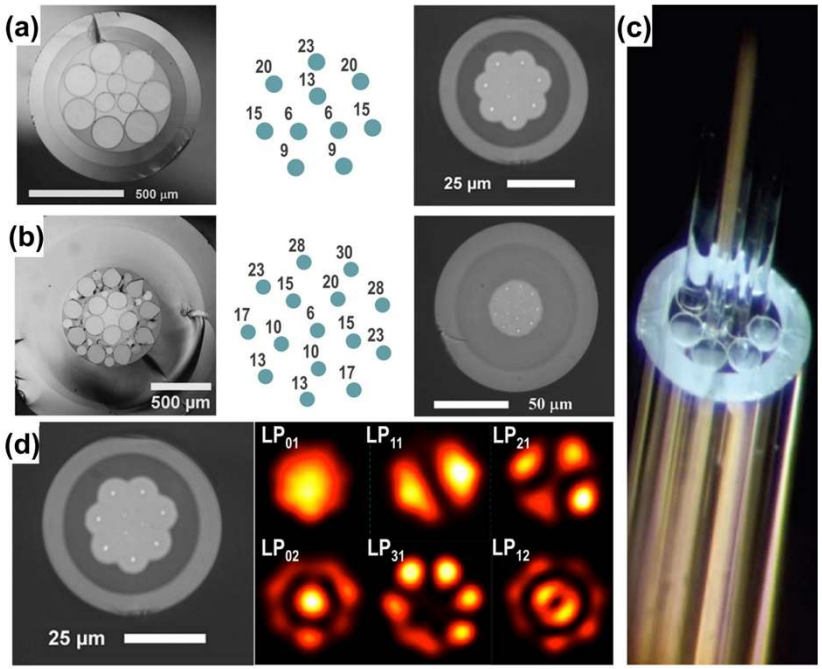
11 Two Smaller Cores

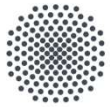


# Mode-Multiplexers

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Nanophotonics 2015; 4:332–352

DE GRUYTER OPEN

**Review Article**

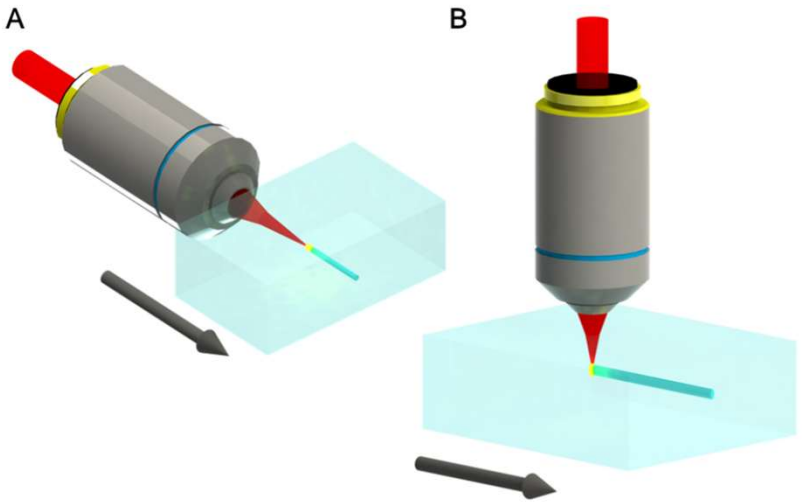
**Open Access**

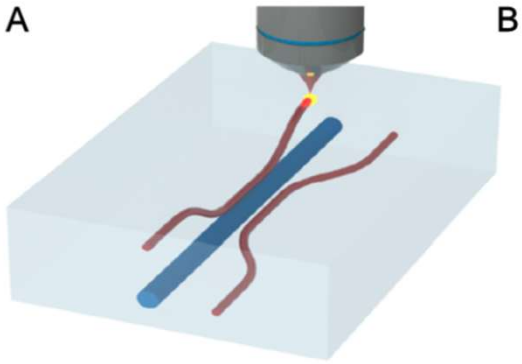
S. Gross* and M. J. Withford

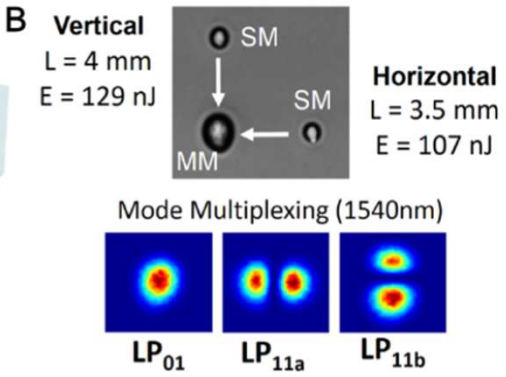
# Ultrafast-laser-inscribed 3D integrated photonics: challenges and emerging applications

# Mode-Multiplexers

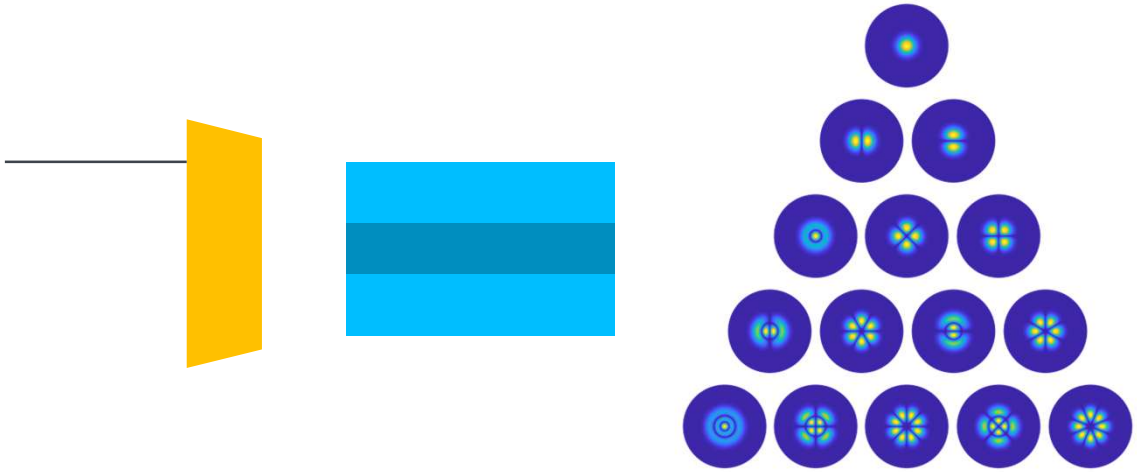
University of Stuttgart
Germany

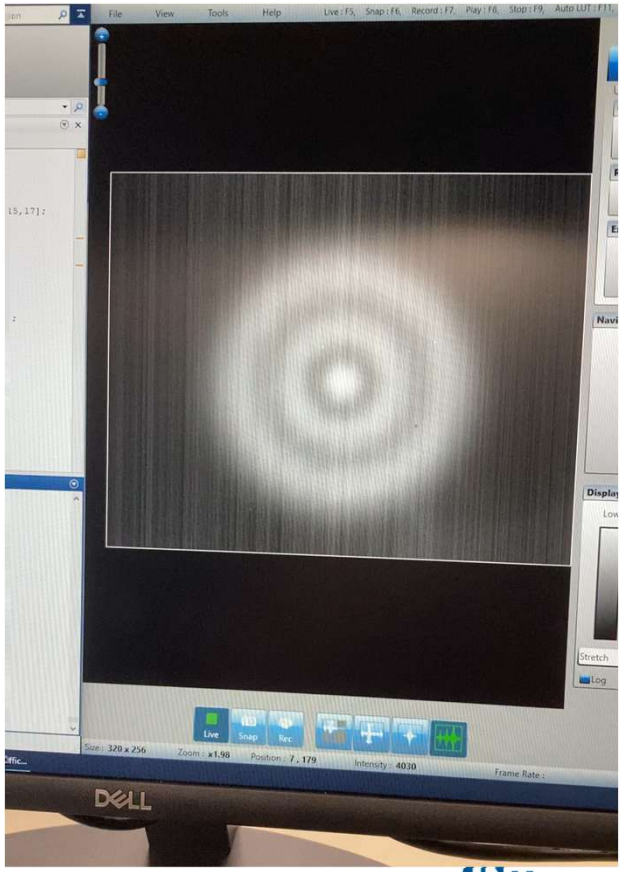


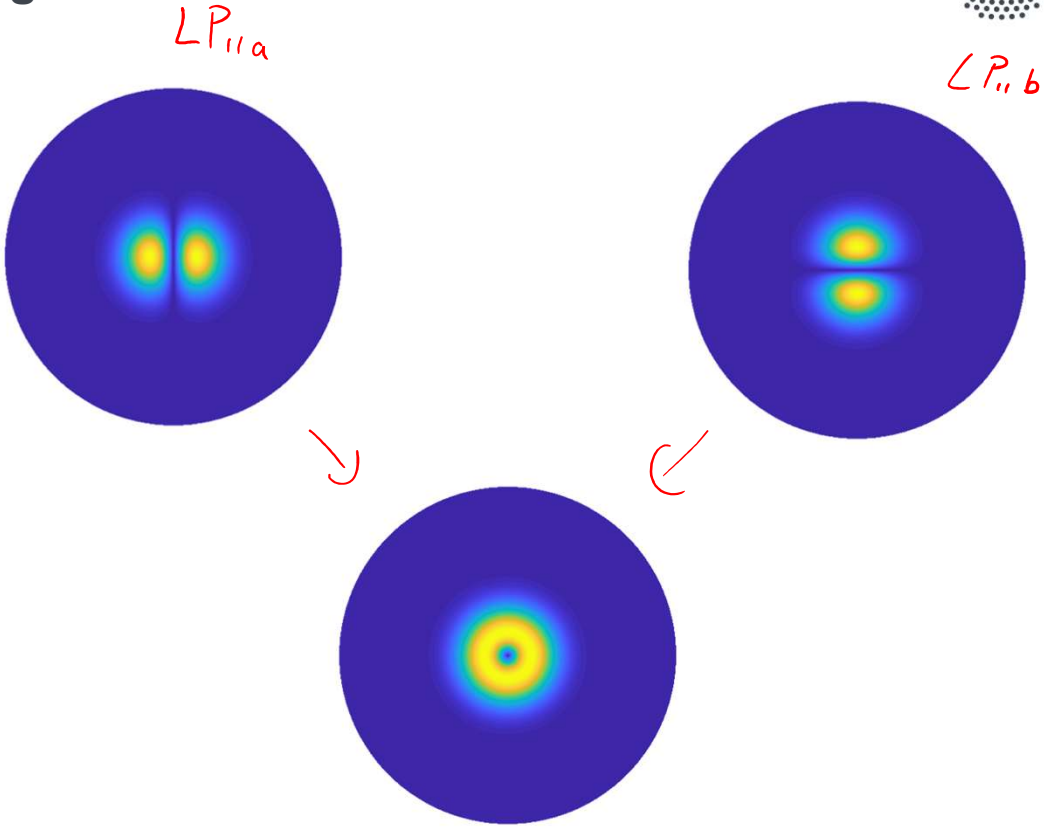



# Graded-index fibers





# Mode Coupling



# Connecting optical fibers:

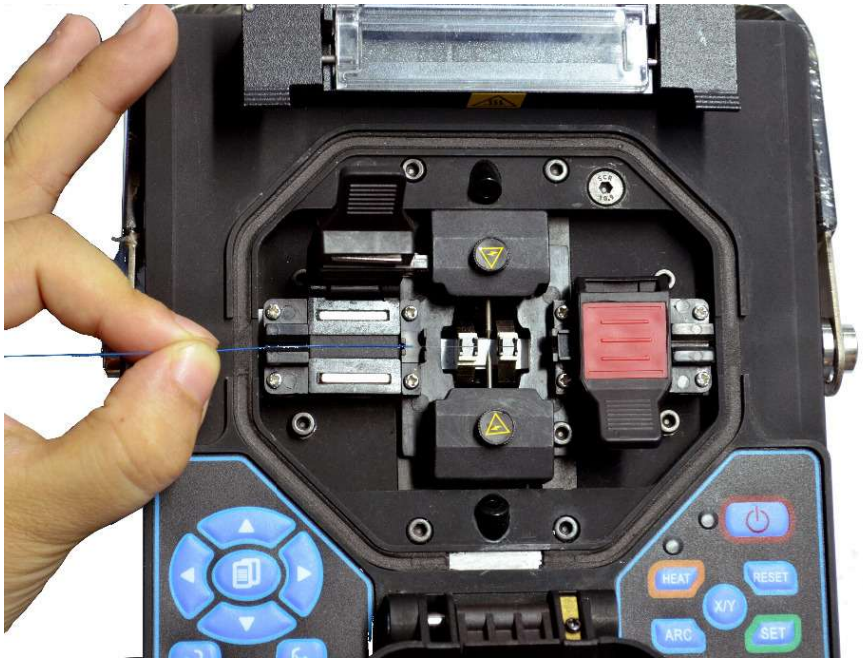
University of Stuttgart
Germany

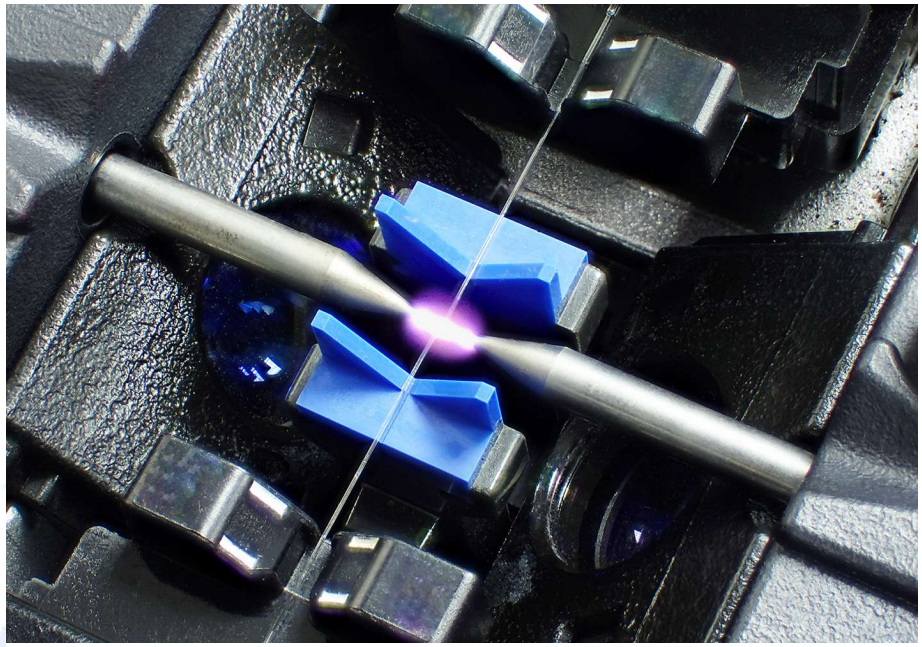




# Connecting optical fibers: fusion splicing

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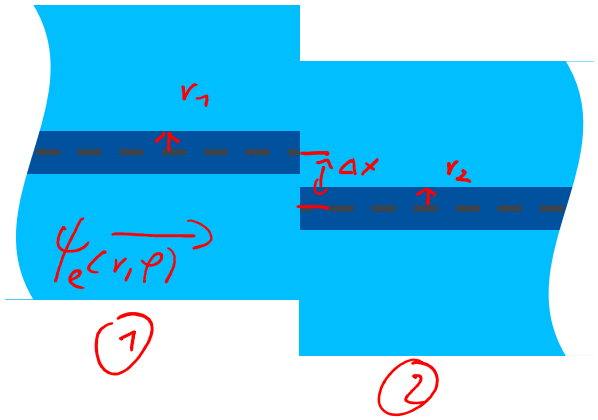


- Loss below 0.1 dB

# Butt-coupling of ~~SI~~ MMFs

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$$\Delta_t \psi_\mu + (n^2(r) - \beta_\mu^2)\,\psi_\mu = 0$$

# Butt-coupling of SI MMFs

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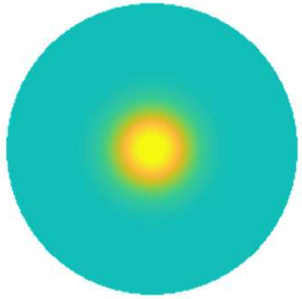
$$\iint_A \psi_\mu \psi_\nu^* \, dA = 0 \qquad \text{for all } \nu \neq \mu$$

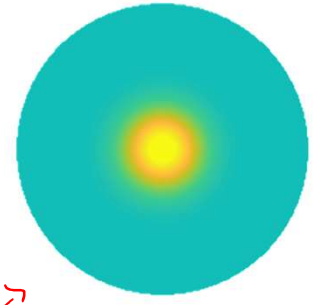
$$\psi_e(r,\varphi) = \sum_\mu c_\mu \psi_\mu(r,\varphi)$$

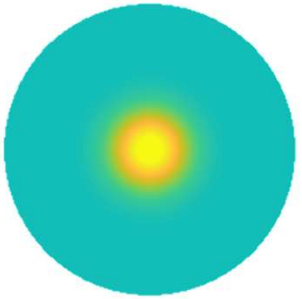
$$\iint_A \psi_e \cdot \psi_\nu^* \, dA = \sum_\mu c_\mu \iint_A \psi_\mu \cdot \psi_\nu^* \, dA \,, \quad c_\mu = \frac{\iint_A \psi_e \psi_\mu^* \, dA}{\iint_A \psi_\mu \cdot \psi_\mu^* \, dA}$$

# Butt-coupling of SI MMFs

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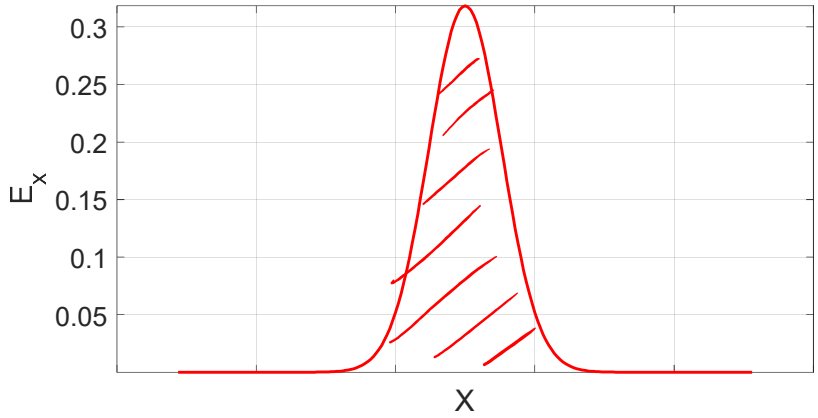




$\psi_e$

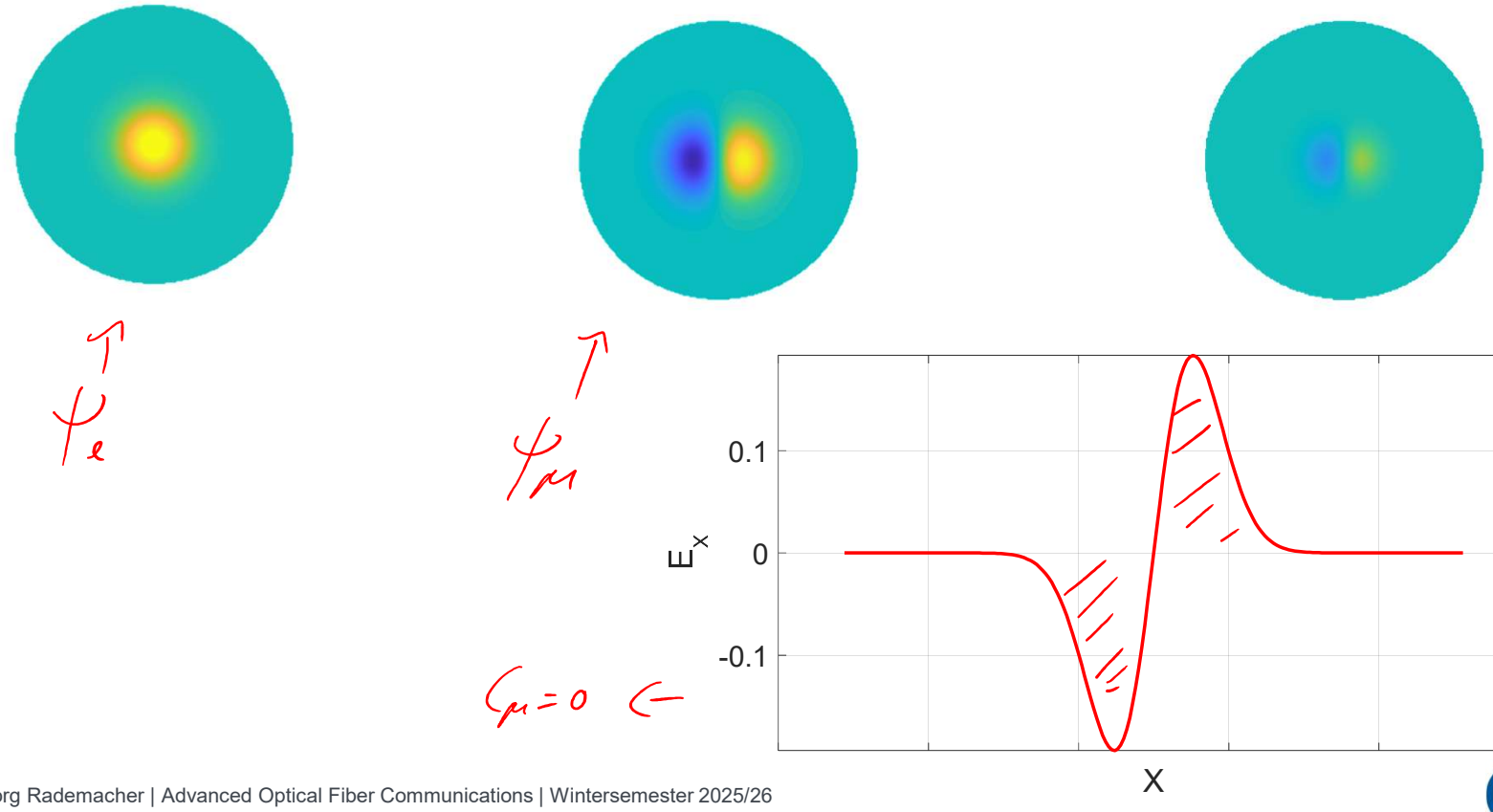
$\psi_\mu$

$$\zeta_\mu = \frac{\iint_A \psi_e \psi_\mu^* \, dA}{\iint_A \psi_\mu \psi_\mu^* \, dA}$$

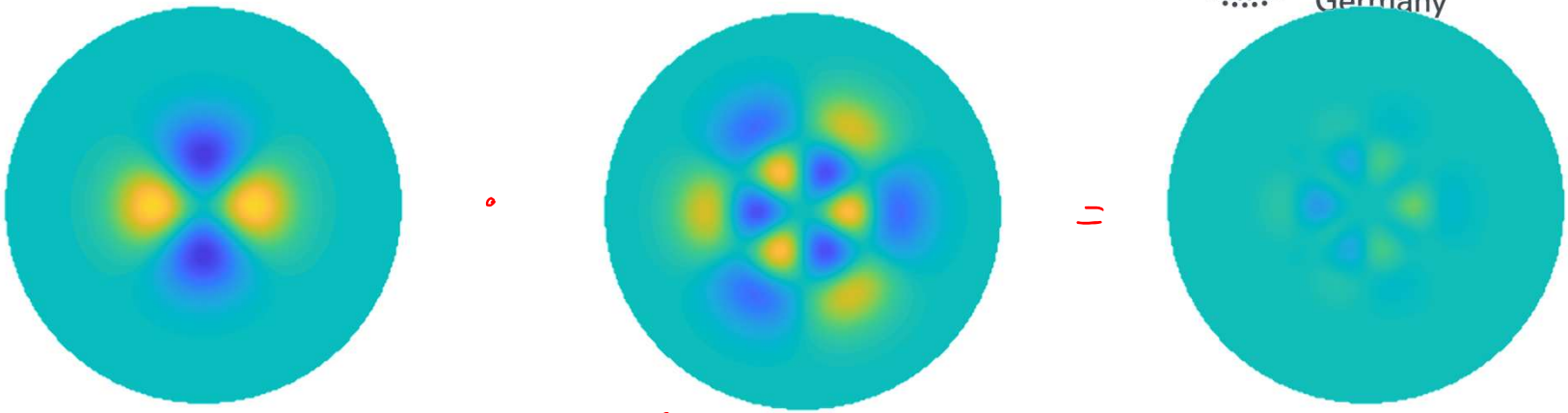


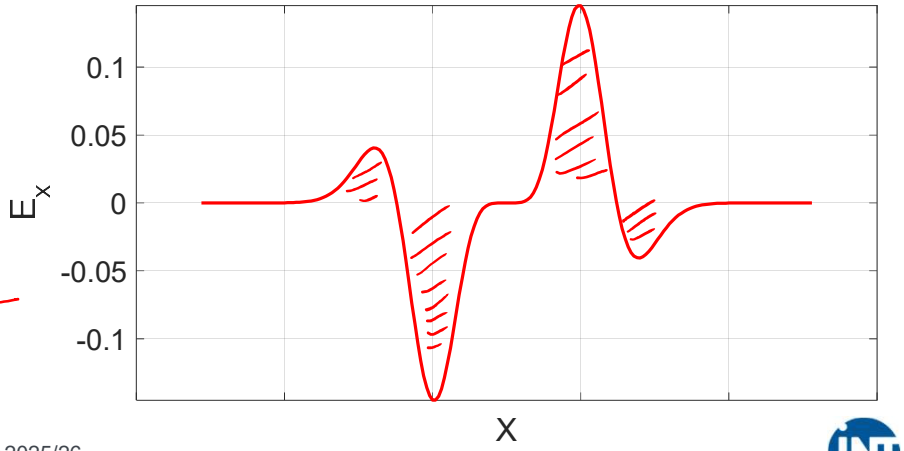
# Butt-coupling of SI MMFs

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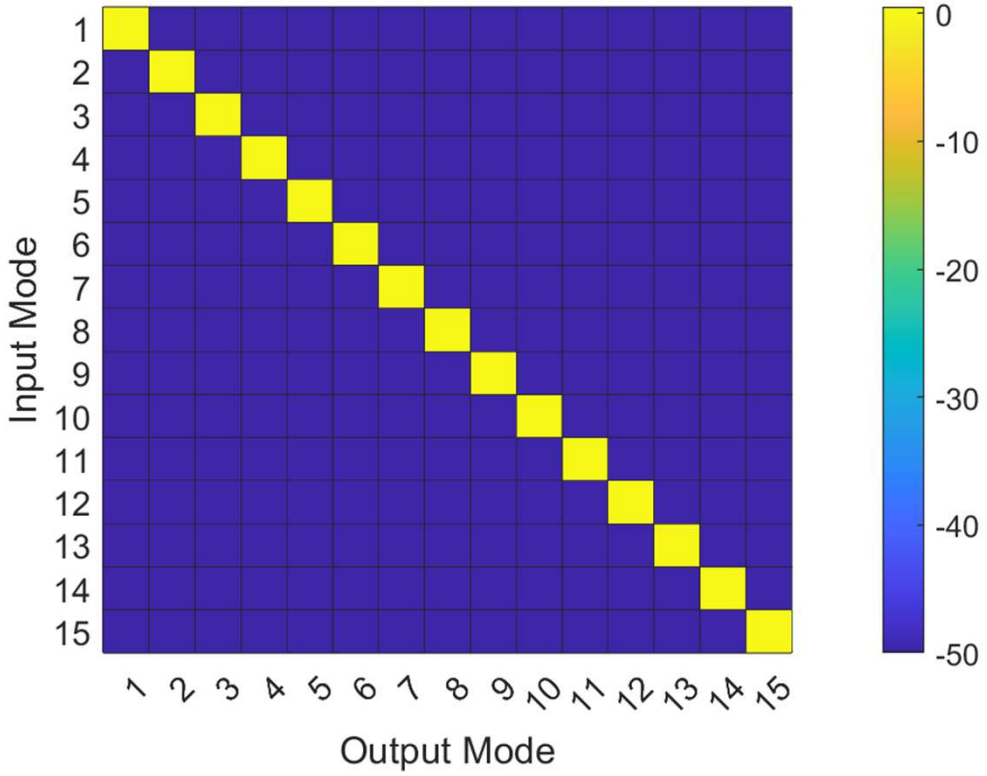
# Butt-coupling of SI MMFs





# Simulation: Coupling coefficients

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$C_\mu$ In dB



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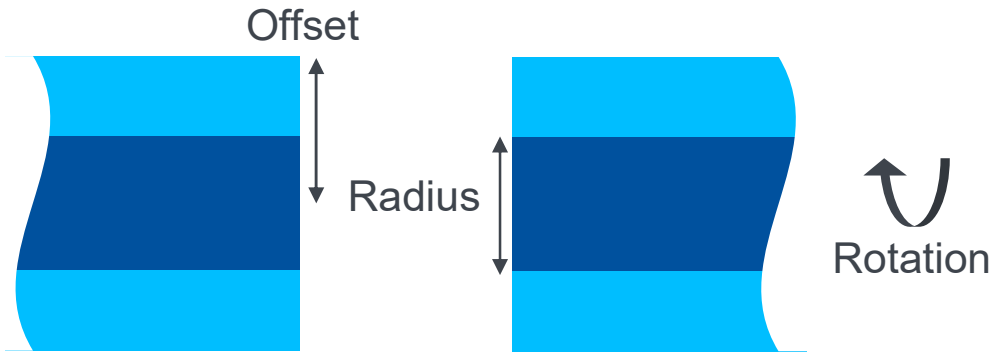
# Concept Test

## Concept test

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Which of the three ways doesn‘t lead to stimulating other modes than the incident mode of the outgoing fiber?

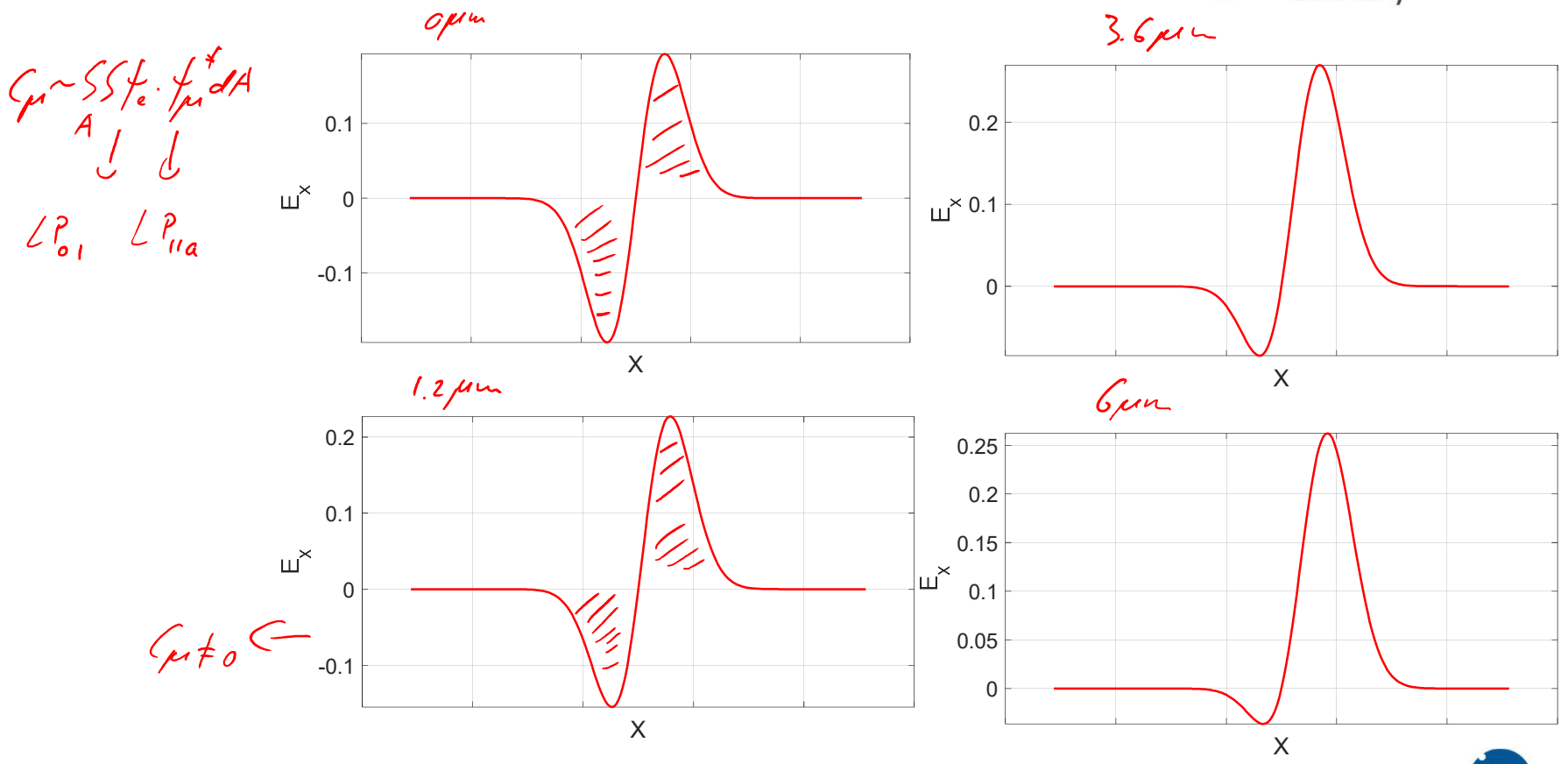
- **A** Different radius
- **B** Offset
- **C** Rotation




https://ilias3.uni-stuttgart.de/vote/DAN1

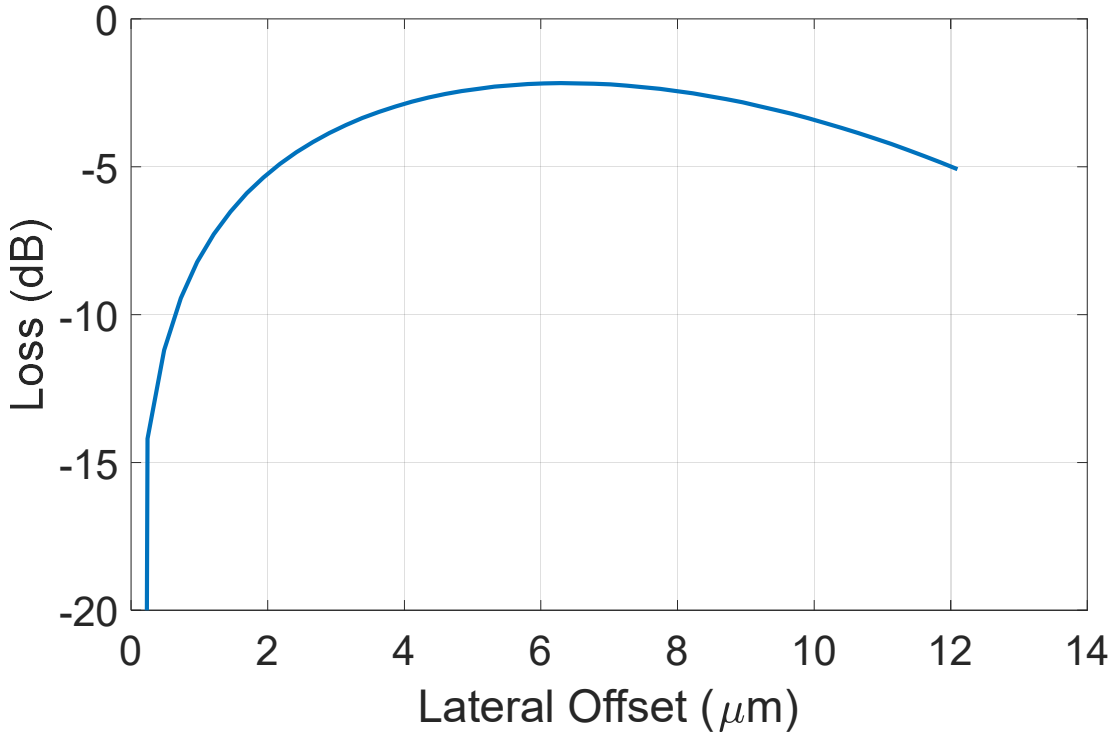
# Butt-coupling of SI MMFs: with lateral offset

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# Butt-coupling of SI MMFs: with lateral offset

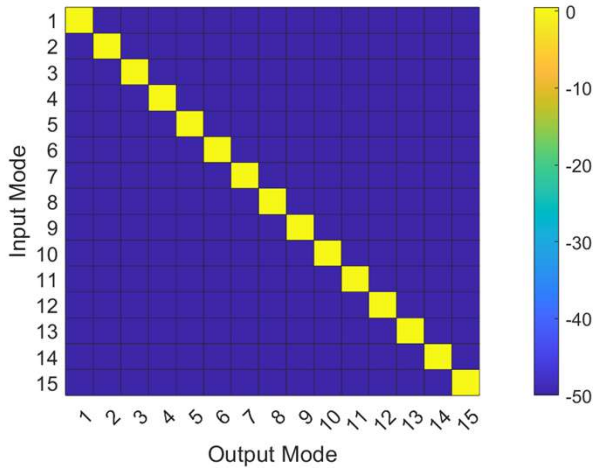
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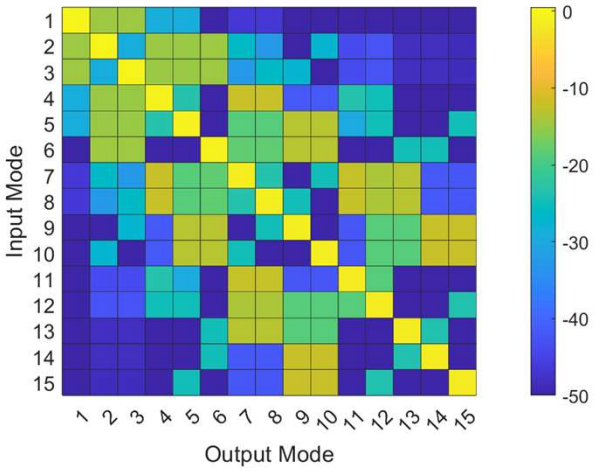
# Simulation: Crosstalk vs. offset

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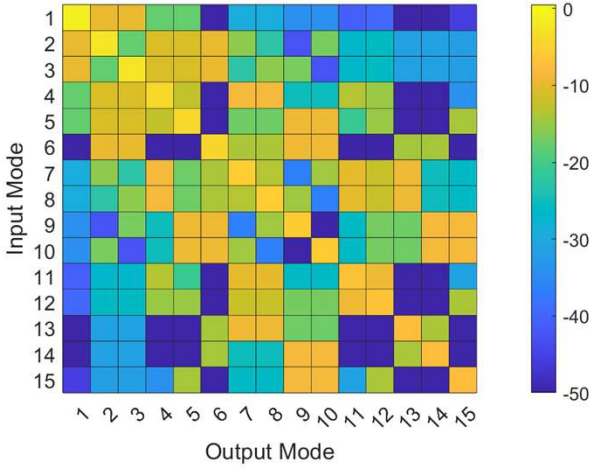
- 0mu m




- 1mu m

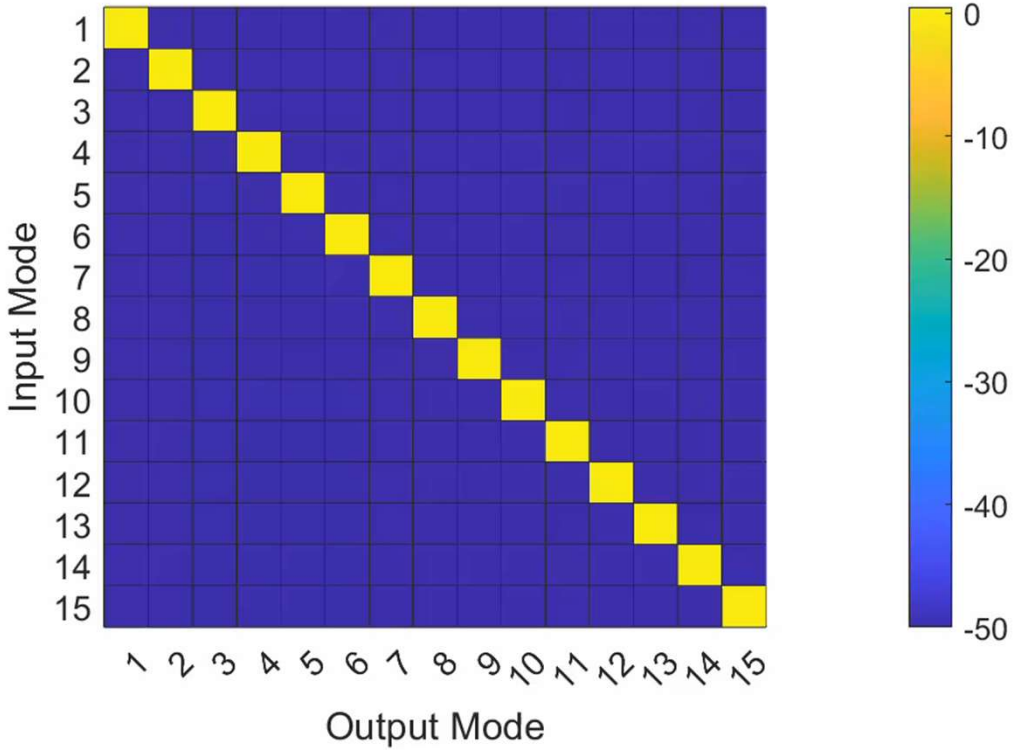



- 3mu m



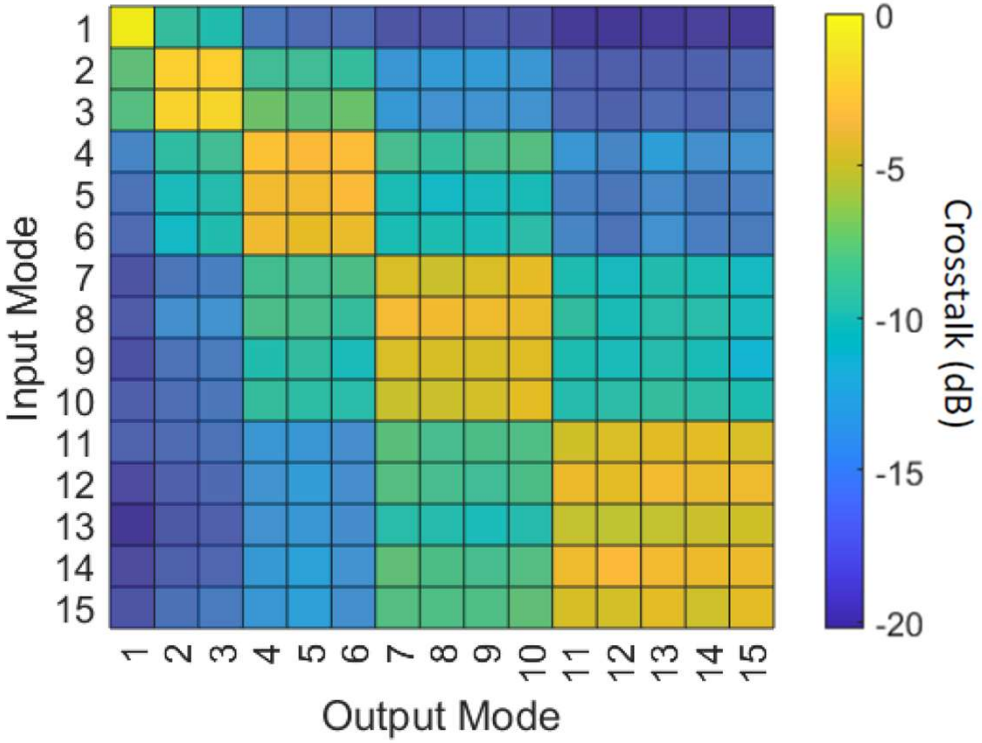
# Simulation: Crosstalk vs. offset

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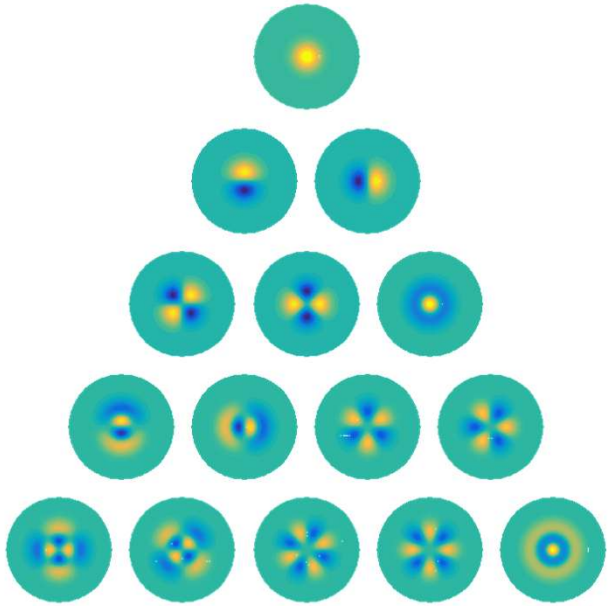





# Experiment: Crosstalk




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# Thank you!



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